CEITEC

MUNI



C9555: A survival guide to simulations

Week 4

Denys Biriukov – Oct. 17th, 2024

Get ready for practical session

- Open new terminal `Ctrl + Alt + T`
- Course materials at: https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555
- Any technical issues/problems/questions? Checklist:
  - Access to Wolf.
  - Proper infinity and conda environments.

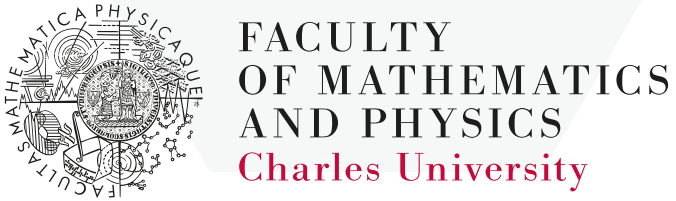
Acknowledgments

This part of the course is inspired and partially based on:

- Teaching materials of “Spring School in Computational Chemistry 2023” taken place in Finland and prepared by Dr. Hector Martinez-Seara and Dr. Matti Javanainen.
- Teaching course about molecular dynamics simulations at Charles University by prof. Lukasz Cwiklik.







More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let’s chat during a break or after

This course is mainly practical – theory will be trying to catch up, not vice versa

Get ready for practical session

- A bit of things to be done:
  - `cd`
  - `mkdir C9555/`
  - `mv ~/survival_guide_simulations_c9555/week*/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week04/`

- `bash ../infinity/infinity-temp.sh`
- `bash ../conda/conda-activate.sh`
- `bash ../conda/jupyter-start.sh`

What is today?

- Week 4:
  How to run MD simulations?
- How and why MD is useful
  - Physics behind MD simulations
- GROMACS
  - Perhaps, the fastest freely-available MD engine
- Jupyter notebooks
  - Easy way to log and reproduce your own work using Python environment
- Matplotlib
  - Quickly plotting data within Python

TEACHING WEEK	DATE	LECTURER	TOPIC
01	26.09	TR	Introduction and visualization with VMD
02	03.10	WM	MD Theory and Lennard-Jones Phases
03	10.10	WM	Lennard Jones Polymers
04	17.10	DB	How to run MD simulations?
05	24.10	DB	Adding salt to our water
06	31.10	TR	Pulling in MD
07	07.11	TR	Enhanced sampling in MD
08	14.11	DB	Membranes in MD
09	21.11	IK	Sampling in MD
10	28.11	IK	Free energies (1/2)
11	05.11	IK	Free energies (2/2)
12	12.12	WM	Using Alpha Fold in MD

Today program

Lecture

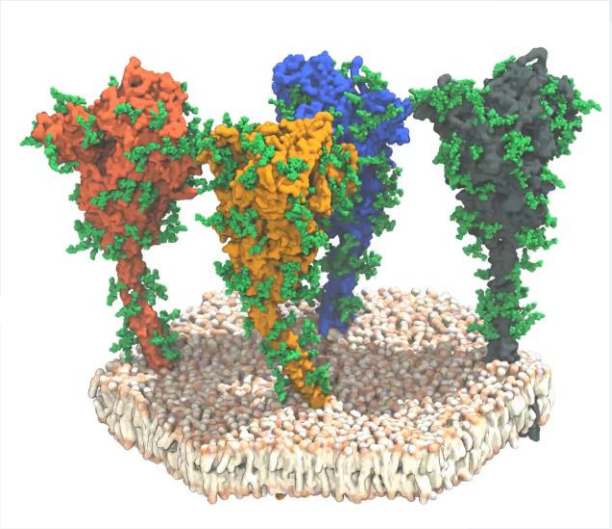
- Recall of basics of MD: why we need it and what physics is behind.
- MD ensembles, relation to thermodynamics.

Practical session

- Jupyter notebook and GROMACS – what is it and how to use it?
- Basic tools to build a system and run a simulation in GROMACS.

Molecular Simulations: Physics, not Cartoon

- Why do we actually use simulations?
  - Visualizing experiments? Reproducing? Confirming?
- A molecular simulation is a well-controlled *in silico* experiment itself!
  - “Computational microscope”, “computational experiment”.
- Good simulation model – predictive power of simulations.
  - Simulations can initiate new experimental work.
  - Certain computational screenings of novel drugs are more efficient (and much cheaper) than many (if not all) experiments.
- Good model = fundamental physics behind it (approximation of fundamental physical laws).

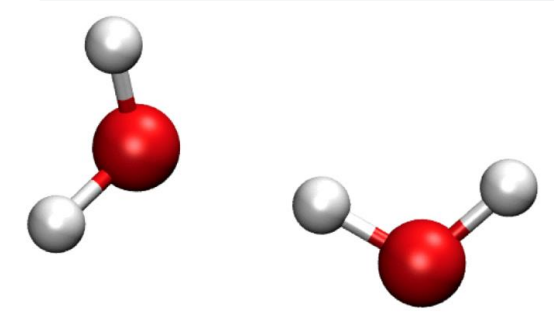


Spike protein of SARS-CoV-2

TUROŇOVÁ, SIKORA, SCHÜRMANN et al.
MAX PLANCK INSTITUTE OF BIOPHYSICS
PAUL-EHRLICH-INSTITUT

Quantum physics?

- Biosimulations usually do not have electrons. Is it legal?
  - Atoms are represented as billiard balls.
  - Averaged effects of electrons are considered through a set of parameters (force fields).
  - Force fields define the potential energy of a system as a function of atomic positions.



- This approximation does not work:
  - Electron transfer processes.
  - Reactive systems.
  - Strong quantum effects.

- Use instead: ab initio molecular dynamics (AIMD), quantum mechanics/molecular mechanics (QM/MM) methods.

Born-Oppenheimer approximation





Born-Oppenheimer approximation





Born-Oppenheimer approximation

Born-Oppenheimer approximation:

- $M_e << M_{nucl}$ ⇒ nuclei move in an averaged field of electrons ⇒ motion of electrons can be separated from motion of nuclei

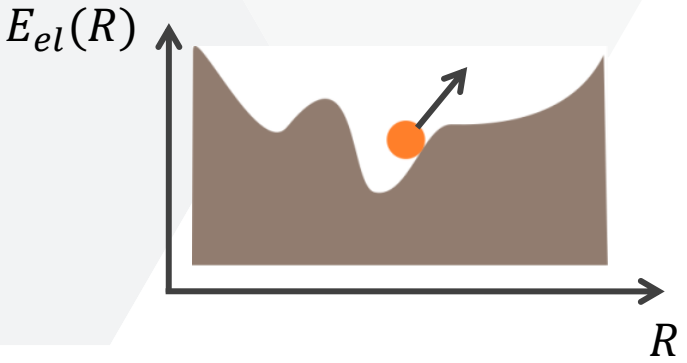
$$\Psi_{tot} = \psi_{el}(r;R)\psi_n(R)$$

for electrons:

$$H_{el}(r;R)\psi_{el}(r;R) = E_{el}(R)\,\psi_{el}(r;R)$$

for nuclei:

$$[T_n + E_{el}(R)]\psi_n(R) = \mathrm{E}\psi_n(R)$$



- Nuclei move on a potential energy surface $E_{el}(R)$, which can be also denoted as $V(\vec{r})$
- System remains in one electronic state.
- Fails: degenerate energy levels.

- Classical mechanics can be used for nuclei
  - de Broglie wavelength (represents a wave-like property of particles) is exceedingly small compared to the size of nuclei
- Not the case of an electron

Born-Oppenheimer approximation

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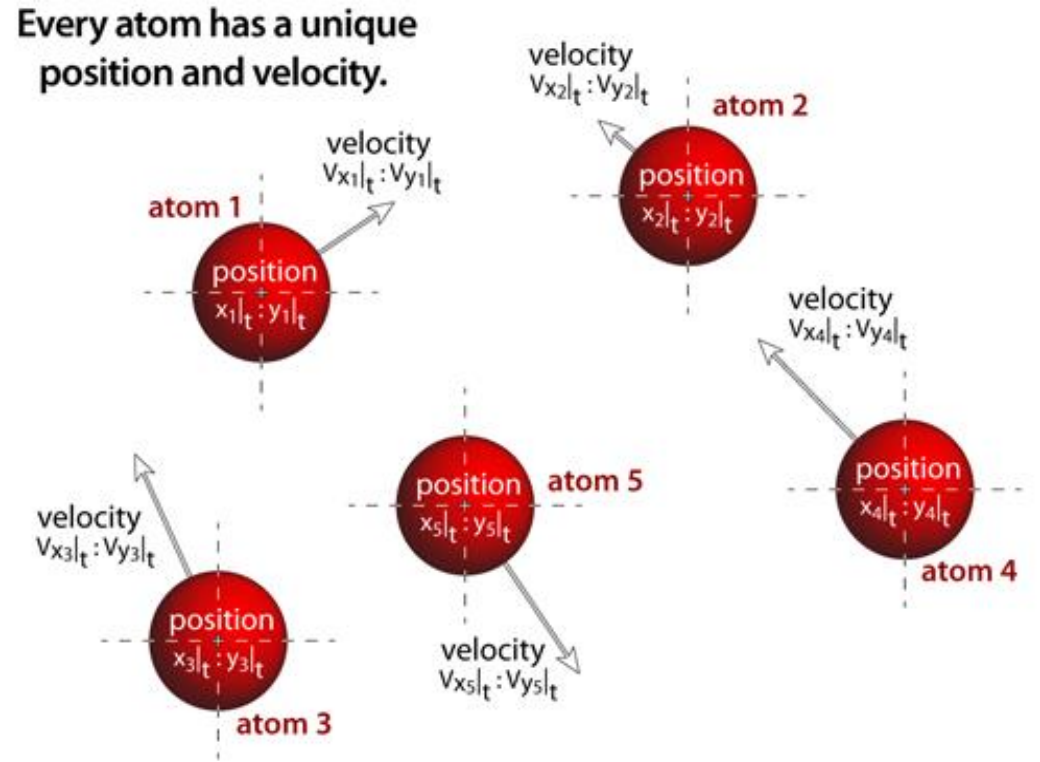
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- $V(\vec{r})$ – force field (a set of predefined parameters) in classical molecular dynamics.
  - **Used in biosimulations!**
- … or can be calculated on-the-fly from the electronic wave function in *ab initio* molecular dynamics simulations.
  - Not used in biosimulations, only small systems possible, computationally very expensive, short simulation times, …

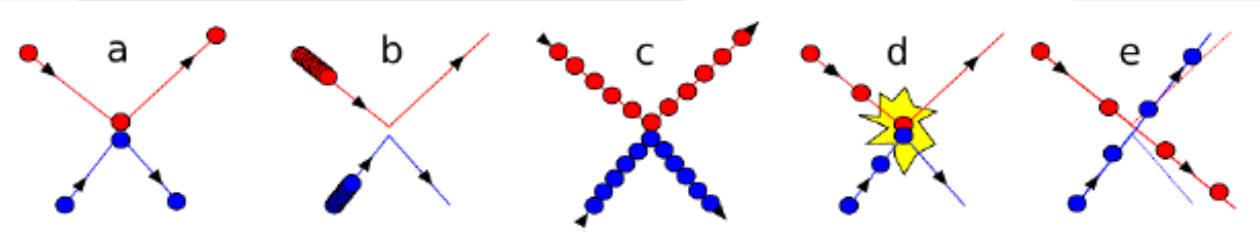
Time evolution



- Every atom has a unique position (in this example, x and y) and velocity (v_x and v_y) .
- They move according to Newton's laws of motion.
- Acceleration ($\vec{a}$) acts on every atom and is determined by the force ($\vec{F}$) acting on the atom divided by the atom's mass (m).

http://atomsinmotion.com/book/chapter5/md

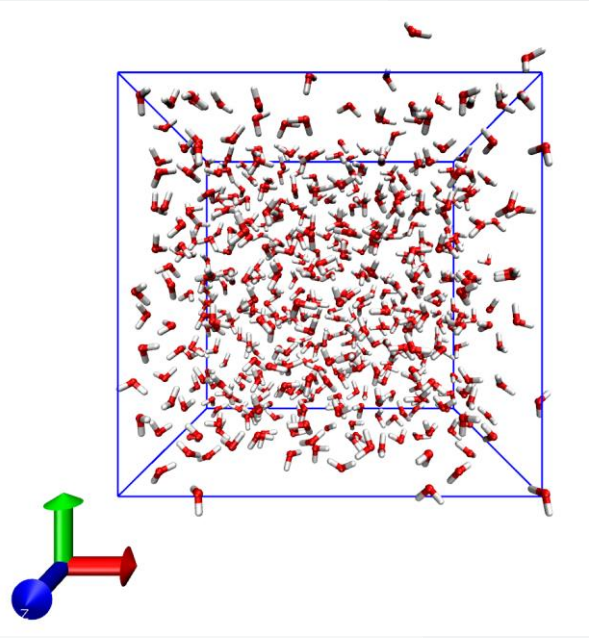
Time step: optimal Δt

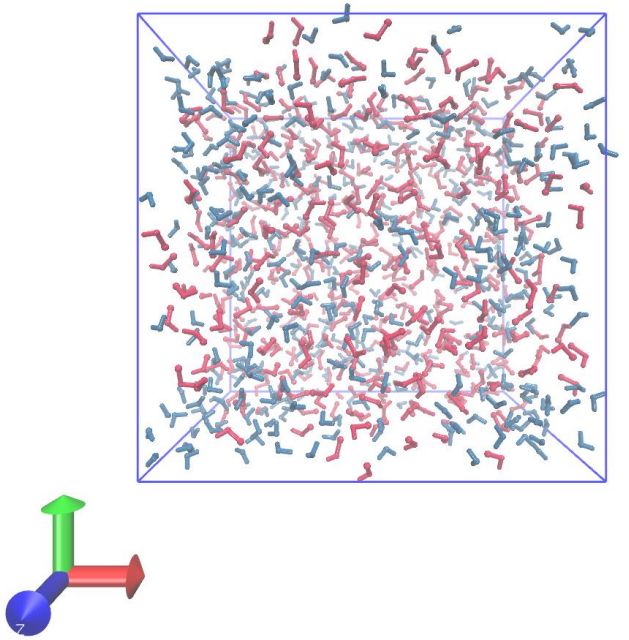


- Time step: not too big, not too small.
- Depends on the force field and its implementation.
- Rule of thumb: at least, one order of magnitude shorter than a characteristic time of the fastest motion in the system.
  - vibration of bonds, time scale ~10^{-14} s, *i.e.*, time step ~1–5 fs

Micro and macro

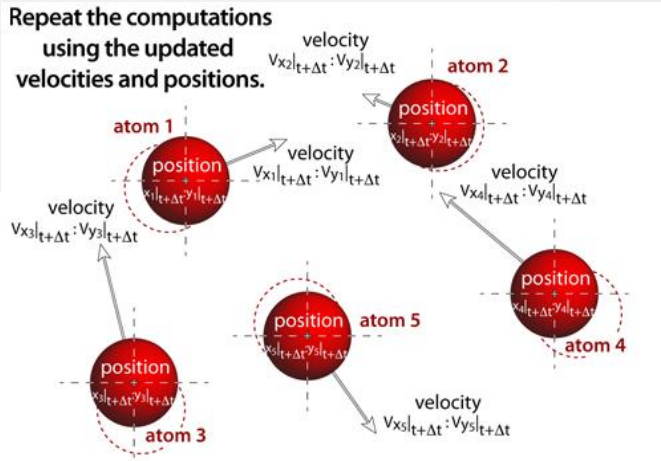
- Newton’s equations of motion govern microscopic evolution (atomic positions, atomic velocities).
- They do not explicitly give macroscopic properties (e.g., temperature, pressure, etc.)
- Macroscopic properties are obtained from trajectories via statistical mechanics (statistical thermodynamics).



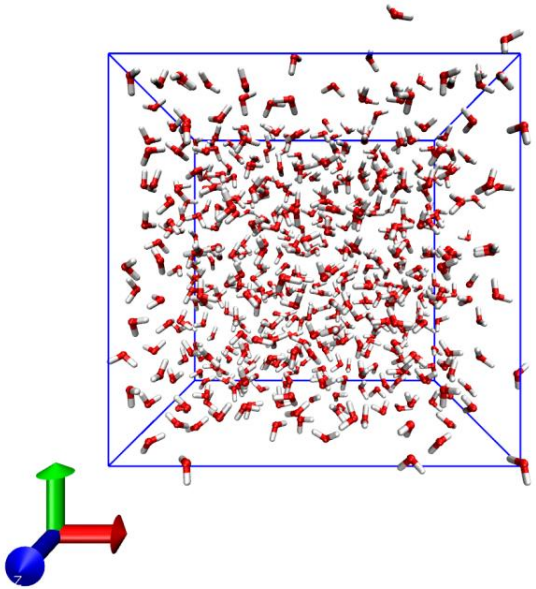


Micro and macro

- Microscopic properties: $r\ (x, y, z)$, $v\ (v_x, v_y, v_z)$



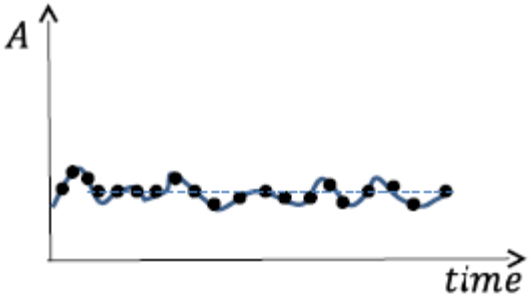
- Macroscopic properties: $T, P, V, \rho, \eta, \ldots$





Ensembles and time

- Ergodic hypothesis: all accessible microstates are equiprobable over a long period of time.
  - Average over ensemble – average over time.
- Macroscopic properties are given by probability distribution within an ensemble.
- In molecular dynamics simulations, we “move” within the ensemble and reproduce its probability distribution.
  - Macroscopic properties can be estimated.
  - Applies to equilibrium molecular dynamics.



$$\langle A \rangle_{time} = \frac{1}{N} \sum_{i=1}^{N} A(t_i)$$

under equilibrium

Equilibrium vs. non-equilibrium

from https://www.easybiologyclass.com

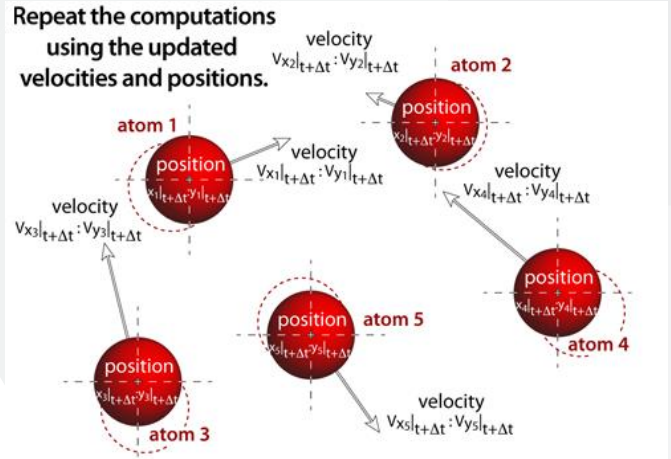


a molecular simulation of ink dissolving in water
is a system out of equilibrium

a molecular simulation of this liquid
is a system under equilibrium

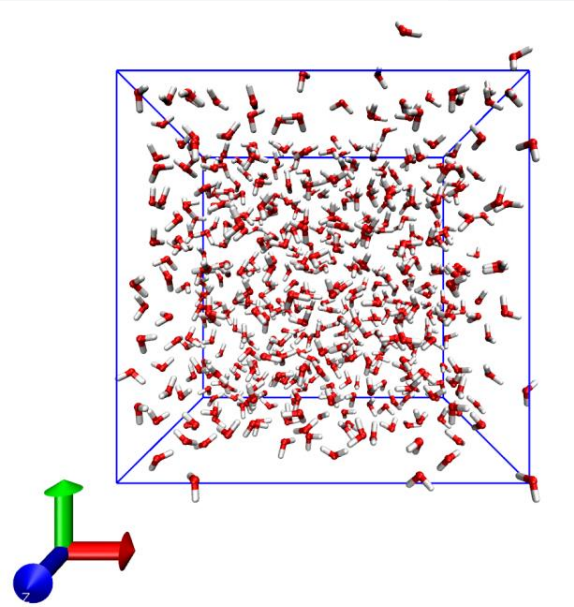
Micro and macro

- If we have velocities, we can calculate kinetic energy and further, e.g., temperature and pressure.



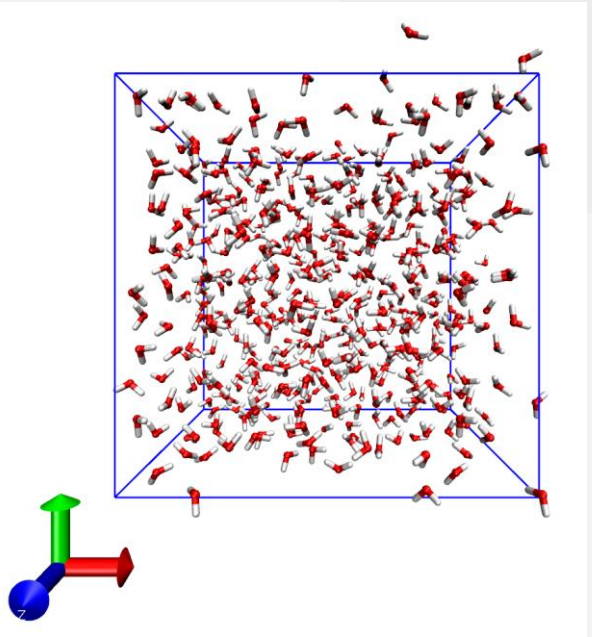
http://atomsinmotion.com/book/chapter5/md

- Relation to pressure:
  - $p = \frac{2N}{3V}\left\langle \frac{1}{2}mv^2 \right\rangle$
- Relation to temperature:
  - $T = \frac{2}{3k_b}\left\langle \frac{1}{2}mv^2 \right\rangle$, k_b – Boltzmann constant



- However, Newton's equations of motion *per se* correspond to the constant energy in the system.
  - Temperature and pressure do not change unless they are adjusted by special algorithms .

Statistical ensembles in MD



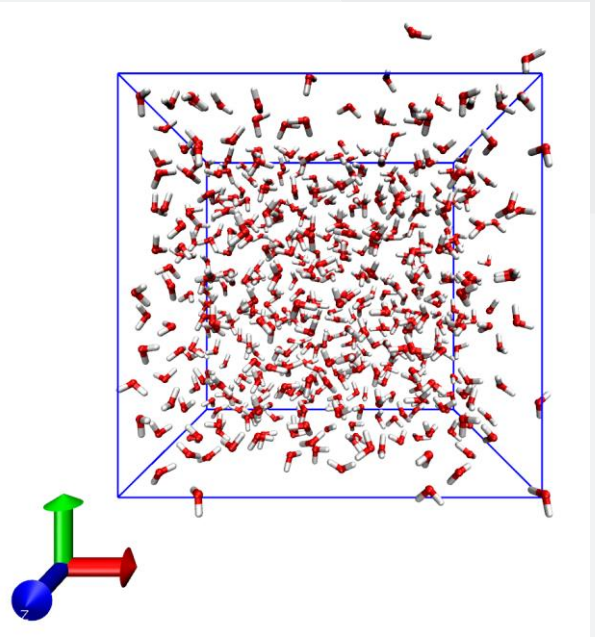
- Typically named by three letters.

- **NVE (microcanonical ensemble):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - E – energy (constant in a simulation).

- Energy is not added nor removed, fixed volume, particles just move in a locked box.
  - Newton's equations correspond to NVE (no external work, constant total energy, no heat exchange).

- Not used in practice: biological simulations are not isolated, big temperature fluctuations, not flexible, convergence and stability issues, etc.

Statistical ensembles in MD



- **NVT (canonical ensemble, isothermal, Helmholtz canonical):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - T – temperature (constant in a simulation).
- Temperature is maintained at a desired value through the exchange of energy with the surroundings (conceptually, of course).
- Sometimes used in bio: simple, stable, realistic enough, excellent for first equilibration stages.
- Disadvantages – volume needs to change in many biosystems, compressibility pressure fluctuations, etc.

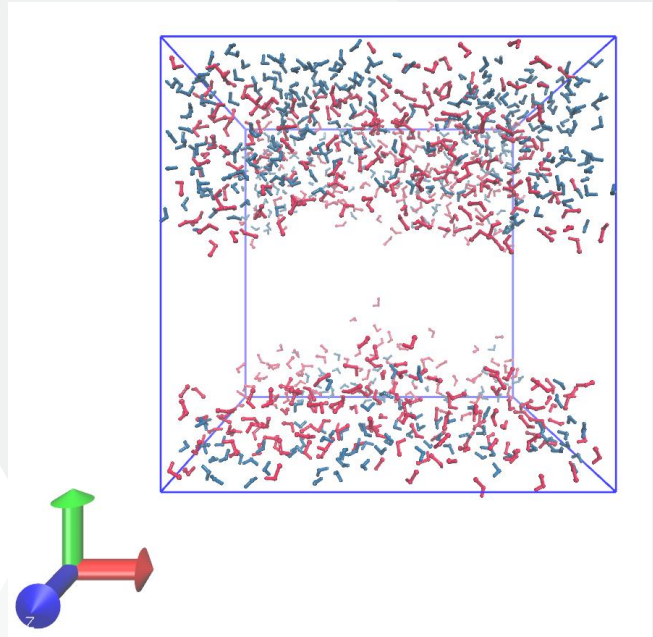
Statistical ensembles in MD

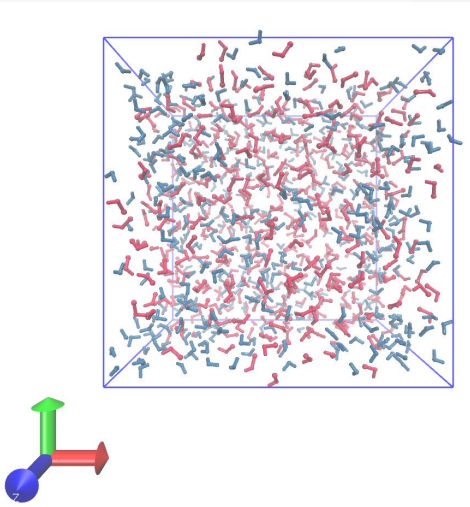
- **NPT (Gibbs canonical ensemble, isobaric-isothermal):**
  - N – number of particles (constant in a simulation).
  - P – pressure (constant in a simulation).
  - T – temperature (constant in a simulation).

- Both temperature and pressure are controlled; pressure is controlled through changing box dimensions.

- 99% use-case in biosimulations, replicates biological conditions, stable, flexible, direct comparison to experimental conditions.

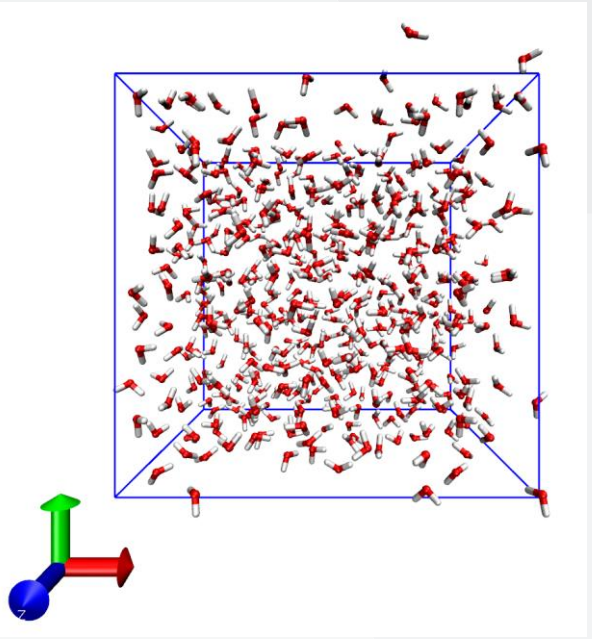
- Disadvantages – basically, technical and computational.





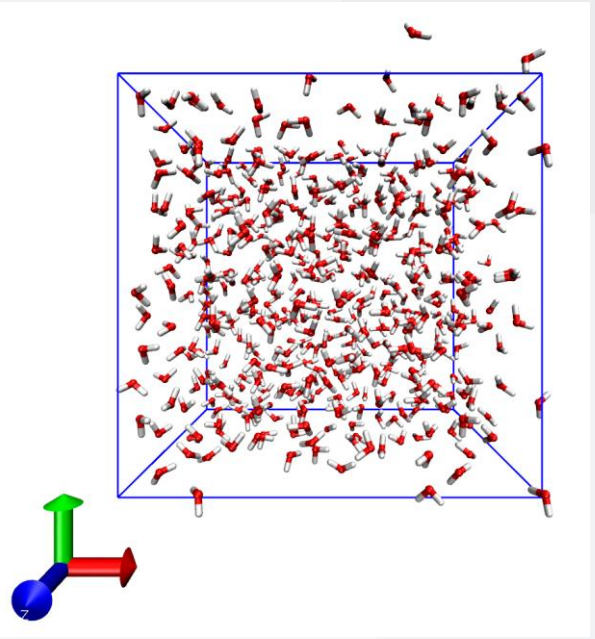
Statistical ensembles in MD

- Other examples:
  - NPH (isoenthalpic-isobaric: number of particles, pressure, enthalpy).
  - NPE (isobaric: number of particles, pressure, energy).
  - μVT (Grand canonical: chemical potential, volume, temperature).
  - μPT (chemical potential, pressure, temperature).
    - μVT and μPT are usually used in Monte-Carlo simulations.
  - NPγT (number of particles, pressure, surface tension, temperature).
  - …
- I think you will never use them.



Bonus: energy minimization

- Used to relax the initial structure, removing steric clashes and unrealistic geometries
- Brings the system to a local energy minimum
- Minimizes possible numerical crashes
- There is no reason why you should not use it



- Today we will test energy minimization, NVT, and NPT

Break!

Get ready for practical session!